

## STRATHMORE AGDM, AB, Canada

WMO: 715260

Lat: 51.0387N

Lon: 113.2897W

Elev: 976

StdP: 90.14

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -27.8      | -24.8 | -31.0                         | 0.2 | -27.7 | -27.8 | 0.3 | -24.7 | 11.5                  | -2.6 | 10.3 | -4.9 | 3.3                   | 210  | 0.680 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.4                   | 28.6            | 16.5 | 26.7 | 15.8 | 25.0 | 15.2 | 18.4                | 25.2 | 17.2 | 24.1 | 16.2 | 23.0 | 3.6                  | 210  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 15.9                            | 12.7 | 21.6 | 14.6 | 11.7 | 20.3 | 13.5 | 10.8 | 19.2 | 56.1          | 25.3 | 52.2 | 24.2 | 49.1 | 23.1 | 24.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 11.0          | 9.4  | 8.3 | -31.5                      | 32.3 | 2.3                | 1.8 | -33.1                                              | 33.6 | -34.5      | 34.6 | -35.7      | 35.6 | -37.4      | 36.8 |    |    |
| (5)               |      |     | -31.7                      | 20.7 | 2.2                | 1.4 | -33.3                                              | 21.7 | -34.6      | 22.5 | -35.8      | 23.3 | -37.4      | 24.3 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 4.3   | -7.1  | -8.4  | -2.4  | 4.2   | 10.6  | 14.3  | 17.4  | 16.1  | 11.7  | 5.1   | -2.6  | -8.1 | (6)  |
| (7)  |                                                                                  | DBStd       | 11.05 | 8.23  | 8.06  | 7.54  | 5.14  | 4.32  | 3.15  | 2.84  | 3.25  | 4.69  | 5.00  | 7.15  | 8.00 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 2798  | 529   | 514   | 384   | 182   | 45    | 3     | 0     | 2     | 34    | 163   | 379   | 562  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 5164  | 788   | 747   | 642   | 425   | 241   | 127   | 51    | 82    | 202   | 411   | 629   | 820  | (9)  |
| (10) |                                                                                  | CDD10.0     | 723   | 0     | 0     | 1     | 7     | 64    | 132   | 231   | 191   | 86    | 11    | 1     | 0    | (10) |
| (11) |                                                                                  | CDD18.3     | 47    | 0     | 0     | 0     | 0     | 1     | 5     | 23    | 14    | 4     | 0     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 764   | 0     | 0     | 0     | 3     | 44    | 82    | 301   | 236   | 95    | 2     | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 161   | 0     | 0     | 0     | 0     | 3     | 13    | 71    | 54    | 20    | 0     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 4.0   | 4.1   | 3.9   | 4.2   | 4.5   | 4.4   | 4.1   | 3.6   | 3.5   | 3.8   | 4.0   | 4.1   | 3.9  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 516   | 19    | 16    | 25    | 38    | 68    | 99    | 73    | 63    | 45    | 26    | 24    | 20   | (15) |
| (16) |                                                                                  | PrecMax     | 678   | 37    | 32    | 54    | 87    | 110   | 254   | 160   | 114   | 110   | 51    | 60    | 44   | (16) |
| (17) |                                                                                  | PrecMin     | 327   | 4     | 3     | 8     | 15    | 24    | 44    | 28    | 12    | 7     | 8     | 5     | 6    | (17) |
| (18) |                                                                                  | PrecStd     | 68    | 8     | 7     | 10    | 17    | 25    | 39    | 31    | 24    | 24    | 11    | 12    | 10   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 8.8   | 9.5   | 15.6  | 22.2  | 27.0  | 28.5  | 31.0  | 30.7  | 29.1  | 22.0  | 14.4  | 10.2 | (19) |
| (20) |                                                                                  |             | MCWB  | 3.5   | 3.8   | 7.1   | 10.1  | 12.8  | 16.7  | 17.8  | 16.0  | 14.6  | 11.2  | 6.7   | 3.2  | (20) |
| (21) |                                                                                  | 2%          | DB    | 6.5   | 6.2   | 12.5  | 18.9  | 24.5  | 25.6  | 28.7  | 28.0  | 26.1  | 18.4  | 11.0  | 6.5  | (21) |
| (22) |                                                                                  |             | MCWB  | 2.4   | 2.1   | 5.2   | 8.6   | 12.1  | 14.8  | 17.5  | 16.1  | 13.7  | 9.6   | 5.0   | 2.6  | (22) |
| (23) |                                                                                  | 5%          | DB    | 5.0   | 4.2   | 9.7   | 15.7  | 22.0  | 23.5  | 26.6  | 26.1  | 23.5  | 15.6  | 8.4   | 4.5  | (23) |
| (24) |                                                                                  |             | MCWB  | 1.6   | 0.8   | 3.8   | 6.9   | 11.0  | 14.1  | 17.3  | 15.7  | 12.9  | 8.2   | 3.4   | 1.1  | (24) |
| (25) |                                                                                  | 10%         | DB    | 3.6   | 2.2   | 7.0   | 12.9  | 19.5  | 21.6  | 24.7  | 23.8  | 20.5  | 13.0  | 6.1   | 2.5  | (25) |
| (26) |                                                                                  |             | MCWB  | 0.7   | -0.4  | 2.6   | 5.5   | 10.0  | 12.9  | 16.4  | 15.1  | 11.8  | 7.0   | 2.0   | -0.3 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 4.3   | 4.1   | 7.7   | 10.8  | 13.9  | 18.0  | 21.0  | 18.7  | 15.8  | 12.1  | 7.4   | 4.5  | (27) |
| (28) |                                                                                  |             | MCD B | 8.2   | 8.2   | 14.2  | 21.2  | 23.7  | 24.8  | 27.0  | 25.7  | 26.1  | 20.5  | 14.0  | 8.2  | (28) |
| (29) |                                                                                  | 2%          | WB    | 2.8   | 2.3   | 5.7   | 9.0   | 12.7  | 16.3  | 19.2  | 17.5  | 14.4  | 10.3  | 5.4   | 2.7  | (29) |
| (30) |                                                                                  |             | MCD B | 6.0   | 6.0   | 11.5  | 17.7  | 22.7  | 23.3  | 25.7  | 24.9  | 23.4  | 17.0  | 10.2  | 6.3  | (30) |
| (31) |                                                                                  | 5%          | WB    | 1.8   | 1.1   | 4.2   | 7.5   | 11.7  | 15.0  | 18.0  | 16.5  | 13.3  | 8.7   | 3.9   | 1.2  | (31) |
| (32) |                                                                                  |             | MCD B | 4.8   | 4.0   | 9.2   | 14.9  | 20.5  | 21.6  | 24.7  | 23.8  | 21.9  | 14.8  | 8.0   | 4.4  | (32) |
| (33) |                                                                                  | 10%         | WB    | 0.6   | -0.4  | 2.7   | 5.9   | 10.6  | 13.9  | 17.0  | 15.5  | 12.2  | 7.3   | 2.3   | -0.3 | (33) |
| (34) |                                                                                  |             | MCD B | 3.5   | 1.9   | 6.6   | 11.9  | 18.1  | 19.9  | 23.5  | 22.6  | 19.9  | 12.4  | 5.6   | 2.4  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 9.4   | 9.3   | 9.8   | 11.5  | 12.3  | 11.2  | 12.4  | 12.5  | 12.4  | 11.2  | 9.6   | 8.9  | (35) |
| (36) |                                                                                  |             | MCDBR | 10.4  | 11.3  | 13.9  | 17.0  | 16.4  | 14.7  | 15.5  | 16.8  | 17.5  | 15.7  | 12.8  | 10.3 | (36) |
| (37) |                                                                                  |             | MCWBR | 7.2   | 8.0   | 8.0   | 8.4   | 6.7   | 6.4   | 7.0   | 7.1   | 7.2   | 8.2   | 8.0   | 7.4  | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 9.9   | 10.9  | 13.3  | 16.1  | 14.9  | 12.8  | 14.0  | 14.8  | 16.1  | 14.9  | 12.6  | 10.4 | (38) |
| (39) |                                                                                  |             | MCWBR | 7.0   | 7.9   | 8.2   | 8.2   | 6.4   | 6.4   | 6.9   | 6.8   | 6.9   | 8.0   | 8.2   | 7.8  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.255 | 0.264 | 0.280 | 0.330 | 0.347 | 0.342 | 0.342 | 0.358 | 0.315 | 0.278 | 0.251 | 0.235 | (40) |      |
| (41) |                                                                                  | taud        | 2.393 | 2.449 | 2.478 | 2.348 | 2.355 | 2.410 | 2.423 | 2.372 | 2.459 | 2.541 | 2.494 | 2.410 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 795   | 889   | 935   | 912   | 907   | 913   | 906   | 871   | 879   | 862   | 803   | 771   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 59    | 74    | 87    | 111   | 116   | 111   | 108   | 108   | 88    | 67    | 53    | 49    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.15  | 2.16  | 3.51  | 4.78  | 5.75  | 6.22  | 6.71  | 5.39  | 3.81  | 2.32  | 1.25  | 0.89  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.09  | 0.19  | 0.28  | 0.35  | 0.39  | 0.38  | 0.33  | 0.37  | 0.38  | 0.27  | 0.11  | 0.08  | (45) |      |

## Historical Trends

|                              |  | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|-------|---------|-------|-----|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)     | (m)     |         |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A     | N/A     |         |
| (47) Regional (78 neighbors) |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A     | N/A     |         |

Nomenclature: See separate page